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Super-resolution reconstruction of terahertz images via a deep learning network with multi-frequency image fusion

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ABSTRACT

Terahertz time-domain spectroscopy imaging (THz-TDS) has demonstrated unique advantages in the field of non-destructive testing. However, the conventional point-scanning imaging approach is often constrained by low imaging efficiency, limited spatial resolution, and inconsistency in multi-frequency image quality. Although existing super-resolution (SR) reconstruction methods can partially improve imaging efficiency, a certain gap still remains between the reconstruction performance and practical testing requirements. To address this issue, we propose an SR reconstruction method (DL-MFIF) that integrates deep learning and measurement-driven multi-frequency image fusion for high-quality and efficient THz-TDS image reconstruction. The proposed method consists of a dual-branch efficient super-resolution transformer (ESRT) network and a genetic algorithm-driven multi-frequency image fusion module. The dual-branch ESRT is designed to leverage joint pre-training and fine-tuning to mitigate the challenge of limited sample size while enhancing the representation of spatial structures and frequency-dependent features. The multi-frequency image fusion module adaptively selects frequency points and optimizes fusion parameters to further enhance image consistency and detail clarity. Experimental results show that, with only 130 THz-TDS point-scan images, DL-MFIF outperformed existing methods in terms of reference-based and no-reference quality indicators. This approach significantly improves the effective spatial detail of reconstructed THz-TDS point-scan images, thereby supporting more efficient and practical THz-TDS measurement applications.

1. Introduction

Terahertz (THz) waves are electromagnetic waves with wavelengths between infrared light and microwaves, with frequencies in the range of approximately 0.1–10 THz. They are characterized by low photon energy, non-ionizing radiation, and strong penetration capability, which provide significant advantages for material safety and tissue compatibility [1]. In recent years, THz imaging technologies have attracted considerable attention in fields such as non-destructive testing (NDT) of non-metallic and non-polar materials, security inspection, and biomedical imaging. Among them, terahertz time-domain spectroscopy (THz-TDS) imaging, which emits broadband THz pulses and records their time-dependent electric field signals, enables the simultaneous acquisition of both structural and spectral information from samples [2]. This technique has become one of the most widely applied imaging modalities.

A typical THz-TDS imaging system generates THz pulses by laser excitation. The pulses propagate through the sample and undergo reflection or transmission, while the receiver records their time-domain responses. The corresponding frequency-domain information is then obtained via Fourier transform to achieve imaging [3,4]. Depending on the detection mode, THz-TDS imaging can be categorized into transmission and reflection configurations, which are suitable for probing internal structures or surface features of low-absorption materials and high-absorption materials, respectively. Among various imaging modalities, point-scan imaging is distinguished by its simple system design and high control precision. Nevertheless, this technique still faces several practical challenges. On one hand, the wavelength of THz radiation is limited by the diffraction effect and is highly susceptible to scattering and absorption by water vapor and airborne particles, which degrade the image signal-to-noise ratio (SNR) [5]. On the other hand, the effective spatial resolution is primarily constrained by the system's point spread function (PSF) and the mechanical scanning step size [6]. As illustrated in Fig. 1(a), reducing the step size to improve effective

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spatial resolution incurs a prohibitive time cost (e.g., 48 min per sample), whereas increasing the step size degrades measurement fidelity, resulting in the loss of geometric details [7]. Furthermore, as shown in Fig. 1(b), the SNR and diffraction effects vary significantly across frequencies, leading to frequency-dependent measurement uncertainty. Therefore, achieving measurement-quality-preserving high spatial detail while maintaining imaging efficiency has become a critical bottleneck limiting the practical deployment of THz-TDS point-scan imaging.

To address the trade-off between measurement efficiency and spatial resolution, researchers have explored solutions from two perspectives: hardware optimization and computational post-processing. Although hardware-level improvements can enhance system performance to some extent, they are often constrained by high costs, long development cycles, and bottlenecks in key component technologies, which restrict their engineering applicability. By contrast, super-resolution (SR) reconstruction methods, as computational post-processing techniques within the measurement chain, offer a promising alternative to improve THz-TDS image clarity and optimize data acquisition efficiency. In recent years, deep learning-based SR approaches have been introduced, leveraging deep neural networks to compensate for details and reconstruct sharper images from low-resolution inputs, thereby mitigating image blurring caused by large step-size sampling [8]. However, applying these data-driven models directly to THz-TDS measurement tasks remains challenging.

First, standard models trained on optical datasets often fail to account for the specific physical degradation mechanisms (e.g., the complex PSF and coherence noise) inherent in THz-TDS, leading to unreliable geometric reconstruction. Second, under point-scan conditions, images at different frequency points exhibit significant differences in sharpness and structural fidelity, while existing SR approaches primarily focus on single-frequency modeling and fail to exploit the complementary information across multiple frequencies. Third, unlike computer vision tasks with abundant data, acquiring high-resolution THz calibration datasets is extremely time-consuming, creating a "small-sample" constraint that limits the reliability of pure data-driven algorithms [9]. In summary, effectively restoring image details and improving resolution under conditions of scarce training data, strong noise, and inconsistent multi-frequency structures remains a critical challenge for SR reconstruction of THz-TDS point-scan images.

To this end, we propose an SR reconstruction framework based on deep learning and multi-frequency image fusion (DL-MFIF) to enhance the reconstruction quality of THz-TDS point-scan images. The method is designed to improve imaging efficiency, image clarity, and model generalization under challenging conditions such as limited training samples, strong noise, and multi-frequency degradation. Specifically, in the first stage, a dual-branch efficient super-resolution transformer (ESRT) network is constructed. One branch is pre-trained on grayscale images from public SR datasets, while the other branch is fine-tuned using a small set of THz-TDS images to adapt its network parameters. A channel adjustment module based on residual connections and dual attention mechanisms is introduced to facilitate deep feature fusion. In the second stage, a genetic algorithm (GA)-based multi-frequency image fusion strategy is employed. By adaptively weighting frequency components based on physical metrics (e.g., entropy and gradient), this module mitigates the impact of stochastic noise and diffraction-induced degradation on reconstructed images, promoting structural consistency across the spectrum. Finally, multiple comparative experiments conducted on THz-TDS point-scan images of photosensitive resin samples demonstrate the effectiveness of the proposed DL-MFIF method in enhancing detail restoration and structural consistency.

The main contributions of this study are summarized as follows: (1) An SR reconstruction workflow for THz-TDS point-scan imaging is proposed to alleviate the coupled trade-off between sampling pitch (scan step size) and effective spatial resolution, enabling large-step sampling while improving spatial detail and cross-frequency consistency under practical NDT conditions. (2) A dual-branch ESRT architecture is introduced, trained via transfer learning using public SR images and a small-scale THz-TDS dataset. Through feature fusion, the model enables efficient detail modeling of THz-TDS images under limited data

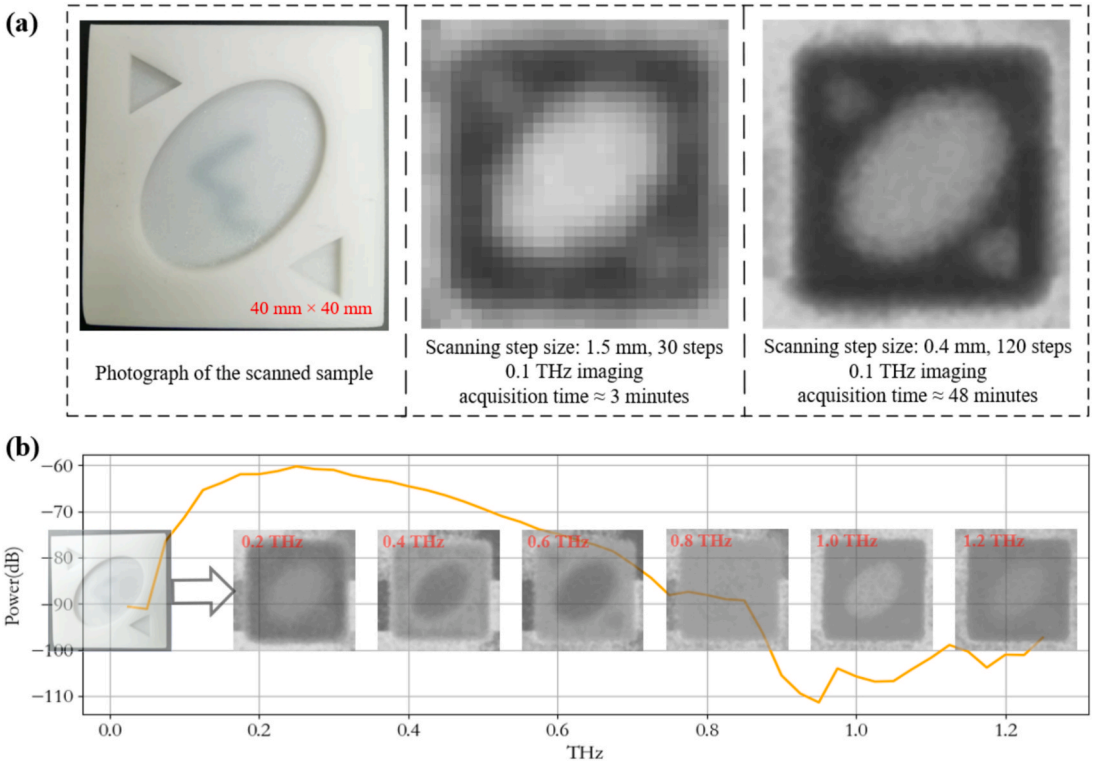


Fig. 1. (a) Imaging comparison of the same sample with different scanning step sizes; (b) Imaging comparison of the same sample at different frequency points.

conditions. (3) A quality-aware GA-based frequency selection and fusion strategy is designed to exploit complementary multi-frequency information while accounting for frequency-dependent SNR and structural fidelity variations, thereby enhancing reconstruction consistency and stability across frequencies.

2. Related work

2.1. Reconstruction techniques

Existing studies on SR reconstruction of THz-TDS images can be broadly categorized into interpolation-based, reconstruction-based, and machine learning-based approaches. Interpolation-based methods [10,11] provide high real-time performance; however, they lack high-resolution priors and are applicable only to scenarios with linear and spatially invariant properties. Due to the complex degradation characteristics of THz-TDS imaging, such methods often result in artifacts and the loss of high-frequency information in the reconstructed images. Reconstruction-based methods place greater emphasis on the degradation process of THz-TDS images caused by factors such as diffraction limits, environmental disturbances, and noise fluctuations. These approaches attempt to recover high-resolution images by constructing inverse models of the imaging degradation process, often incorporating spatial priors to enhance reconstruction stability. For example, Wang et al. [12] introduced a Laplacian operator to constrain image gradients and extended deconvolution into the complex domain, thereby effectively mitigated artifacts in THz-TDS images. Although such methods achieve higher reconstruction accuracy compared with interpolation-based approaches, they are prone to overfitting when training data is limited and often suffer from slow convergence, high computational complexity, and insufficient edge structure recovery.

Machine learning-based SR reconstruction methods can be further divided into traditional machine learning approaches and deep learning approaches. In recent years, research on traditional machine learning methods have been relatively limited. For instance, Lei et al. [6] proposed a local pixel-based graph neural network (GNN) that extracts Fourier graph structures with rotation and flip invariance from low-resolution images. By leveraging graph neural mechanisms to learn the mapping between graph states and target pixels, this approach enhanced the resolution of THz-TDS images. Although such methods demonstrated a certain degree of adaptivity and relatively low computational cost, they still suffered from long processing times, blurred image edges, and the need for parameter tuning across different datasets or imaging devices. With the remarkable success of deep learning in natural image SR tasks, new opportunities have emerged for SR reconstruction in THz-TDS imaging. For example, Li et al. [13] designed a multi-scale residual network (MSRN) for natural images, which was shown to enhance edge and texture reconstruction by fusing features across multiple scales. Tai et al. [14] further proposed the deep recursive residual network (DRRN), which leveraged recursive structures and residual learning strategies to achieve high-quality reconstruction with relatively few parameters. Beyond conventional convolutional neural network architectures, diffusion probabilistic models have recently demonstrated strong generative capabilities in image SR tasks. Li et al. [15] introduced SRDiff, the first diffusion model applied to single-image SR, which progressively guided noise toward high-resolution image sampling and achieved superior performance in complex textured scenes. Liu et al. [16] proposed the DDFSRM model, which incorporated a denoising diffusion fusion strategy for line-scan images and effectively improved structural fidelity in low-SNR scenarios. These studies provided valuable insights for SR reconstruction in THz imaging. Focusing on THz-TDS images, Wu et al. [17] addressed common issues such as edge blurring and detail loss by designing an SR reconstruction framework that integrated Transformer and generative adversarial network (GAN) components. The Transformer enhanced long-range dependency modeling, while the discriminator refined texture detail reconstruction, achieving promising results in THz-TDS SR tasks. Similarly, Yang et al. [18] targeted the degradation blur and high-frequency detail loss in THz-TDS images by proposing a deep learning-based reconstruction network that integrated a residual channel attention mechanism. This method leveraged residual learning to stabilize the modeling process with low-frequency information and employed channel attention to reweight feature channels, thereby strengthening the model's response to critical texture and edge details. Nevertheless, existing deep learning approaches for THz-TDS SR reconstruction still face several challenges. First, the scarcity of high-quality annotated THz-TDS datasets hampers the training and generalization of supervised models. Second, many methods adopt paradigms directly from natural image processing, overlooking the unique physical mechanisms and frequency responses of THz-TDS imaging. Finally, most current approaches focus on single-image modeling, with limited exploration of leveraging complementary information across multiple frequency images.

2.2. Multi-frequency image fusion

Compared with approaches that rely solely on single-frequency modeling, incorporating multi-frequency imaging information can further enhance structural representation and detail clarity. A notable feature of THz-TDS imaging is its inherent multi-frequency capability. Each frequency point reveals distinct yet complementary image characteristics: low-frequency images exhibit stronger penetration and are better suited for revealing deeper sample structures, whereas high-frequency images provide higher resolution and are more effective at capturing fine surface details. Therefore, the rational fusion of information across different frequency points holds great potential to significantly improve structural clarity and detail preservation in reconstructed images. At present, multi-frequency information fusion has been extensively studied in fields such as ground-penetrating radar (GPR), medical image reconstruction, and remote sensing, particularly for enhancing structural representation under complex backgrounds. Early research primarily employed signal processing and time–frequency analysis techniques to achieve multi-frequency data fusion. For example, Xiao et al. [19] proposed a multi-frequency GPR signal fusion method based on forward and inverse S-transform to improve the detection accuracy of railway subgrade defects. Xu et al. [20] developed a frequency fusion approach based on weighted superposition and spectral correction, which emphasized maintaining structural consistency across frequencies in the spectral domain, effectively suppressing reflection errors and frequency drift in GPR imaging. Subsequent studies [21] proposed a multi-frequency centroid fusion strategy for time-domain GPR sequences, which integrated data from multiple central frequencies to enhance subsurface structural inversion capability. Ye et al. [22] further extended multi-frequency fusion concepts to ghost imaging tasks, where the superposition of interference patterns at different frequencies was used to improve imaging quality under low-sampling conditions.

To further improve the adaptability and intelligent capabilities of multi-frequency fusion, researchers have introduced statistical optimization and intelligent algorithms. For instance, Bi et al. [23] improved edge imaging performance through weighted averaging of multi-frequency signals. Lu et al. [24] proposed a multi-frequency and multi-attribute GPR data fusion method based on two-dimensional wavelet transform. By incorporating a dynamic weighting mechanism driven by edge detection algorithms, this approach effectively combined the high-resolution advantage of high-frequency signals with the deep penetration capability of low-frequency signals. Shen et al. [25] developed a multi-frequency data fusion method based on joint weighted deconvolution. Building upon conventional frequency fusion, their method introduced a sub-band weighting scheme that not only improved spatial resolution but also suppressed blurring and artifacts caused by attenuation differences among frequencies. Zhao et al. [26] designed a GA-based multi-frequency image fusion strategy, in which

detection accuracy served as the fitness function to automatically optimize the weight distribution of frequency images. This approach effectively overcame the subjectivity and limitations of traditional manual weighting.

In recent years, deep learning has provided new avenues for multi-frequency image fusion. Some studies employed attention mechanisms to guide models in automatically focusing on more discriminative frequency channels, while others utilized Transformers or multi-branch network architectures to achieve joint modeling in the spatial–frequency domain. For example, Zou et al. proposed the MMR-Mamba framework, which integrated the Mamba architecture with a spatial–-frequency fusion mechanism to achieve high-quality reconstruction of multimodal magnetic resonance imaging (MRI) data [27]. Jin et al. proposed a multi-focus image fusion method that integrated deep features with frequency information. This approach employed the Mamba architecture as the backbone for feature modeling and introduced a joint frequency–depth attention mechanism to achieve dynamic information interaction across different frequency channels [28]. These studies highlighted the role of frequency–spatial distribution information in enhancing image modeling capabilities and validated the effectiveness of fusion strategies in low-SNR and structurally complex scenarios. Similarly, Tian et al. employed the SFCF-Net architecture in electrical impedance tomography, constructing a multi-path spatial–frequency channel fusion module to achieve stable reconstruction under complex frequency conditions [29]. Cao et al. applied multi-scale and multi-frequency feature fusion to remote sensing image segmentation, employed a Vision Mamba framework to jointly model semantic information across frequency images, thereby significantly improved recognition accuracy in complex scenes [30].

As an important approach to enhancing image quality, multi-frequency image fusion has demonstrated broad application prospects and significant effectiveness across various domains. However, research in the context of THz-TDS point-scan imaging remains relatively limited, particularly in terms of strategies tailored for small-sample scenarios, non-ideal imaging conditions, and automation requirements. A key challenge lies in fully exploiting the complementary characteristics among frequency images while preserving structural clarity and fine details, and at the same time suppressing redundant information and noise interference. To address this issue, this study leverages the optimization potential of GA in multi-frequency fusion and proposes a fusion mechanism that integrates adaptive frequency weighting, image enhancement, and denoising operations, aiming to achieve high-quality SR reconstruction of THz-TDS point-scan images.

3. Method

As illustrated in Fig. 2, the SR reconstruction process of the proposed DL-MFIF framework consists of two main stages. In the first stage, a dual-branch ESRT module (SR-DESRT) and a channel adjustment module based on ResBlockCBAM (CA-ResBlockCBAM) are constructed. The output feature maps from the two ESRT branches are concatenated along the channel dimension and subsequently processed by the CA-ResBlockCBAM module to achieve deep feature fusion and dimensional alignment, yielding single-frequency SR images as the output of this stage. In the second stage, a GA-based multi-frequency image fusion (MFIF-GA) module is developed to leverage the multi-frequency imaging characteristics of the THz-TDS system for adaptive fusion and reconstruction enhancement. The module takes as input the frequency-point images generated in the first stage. By automatically optimizing the fusion weights, denoising parameters, and contrast adjustment coefficients through GA, a reconstructed image with improved structural consistency is ultimately generated.

This two-stage design is motivated by the practical constraints of THz point-scanning imaging. The first stage focuses on enhancing spatial structural representations from sparsely sampled single-frequency images, which is a high-dimensional nonlinear mapping problem that can benefit from transformer-based models due to their ability to capture long-range dependencies and global spatial context. In contrast, the second-stage multi-frequency fusion is essentially a weight allocation and parameter selection problem guided by measurement-driven criteria (e.g., image sharpness and noise robustness). If a fully learnable fusion network were adopted, it would require substantially more

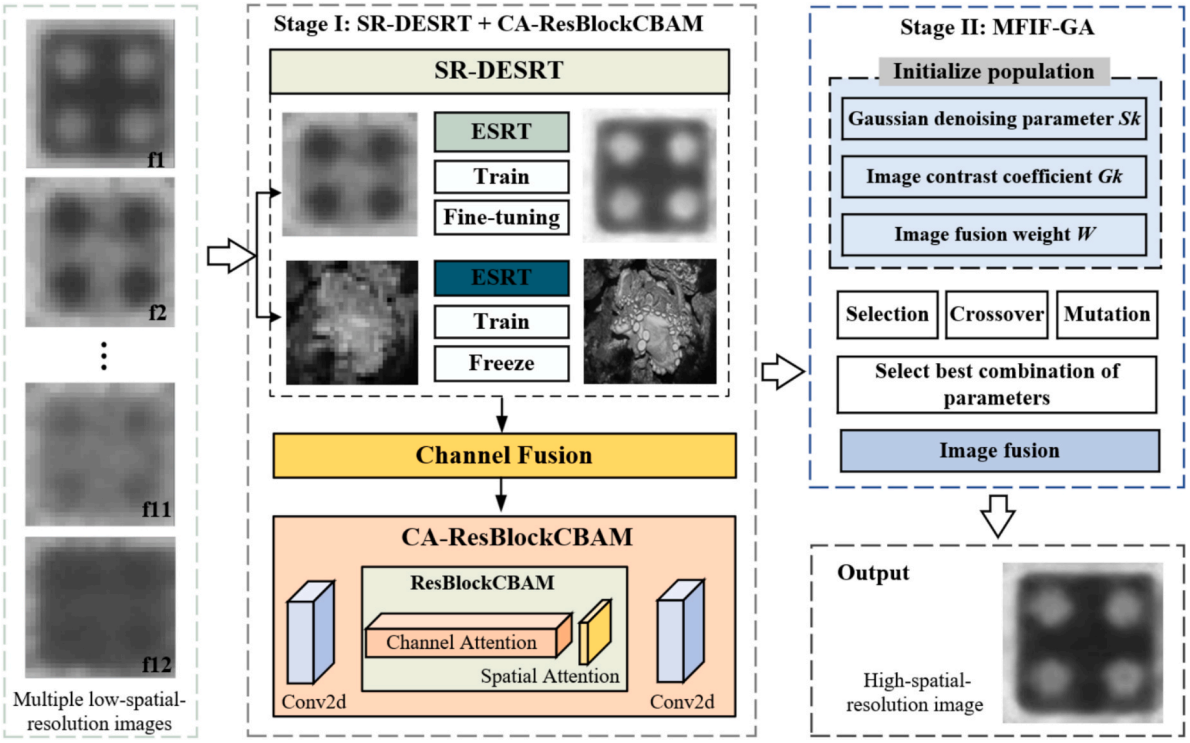


Fig. 2. Framework of the DL-MFIF method.

training data and introduce additional trainable parameters, increasing the risk of overfitting under the current small-sample condition. By comparison, the genetic algorithm, as a global optimization method, can directly optimize measurement-driven evaluation metrics without relying on gradient information, offering improved robustness and controllability under small-sample conditions. Therefore, combining deep-learning-based spatial reconstruction with heuristic optimization for multi-frequency fusion provides a stable and practical solution under realistic THz measurement constraints.

3.1. SR reconstruction based on deep learning networks

3.1.1. ESRT network

In the single-image SR reconstruction task of this study, the ESRT proposed by Lu et al. [31] was employed as the core structure. The ESRT integrates the advantages of lightweight CNNs and Transformers, utilizing a modular hierarchical architecture that significantly reduces computational complexity while maintaining reconstruction accuracy. Specifically, ESRT incorporates a multi-scale feature extraction mechanism. A shallow 3×3 convolutional layer was first applied to capture low-frequency contour features, while the High-Preservation Block (HPB) within the Lightweight Convolutional Block (LCB) dynamically compresses feature maps, preserving high-frequency details while reducing computation. The HPB integrates a high-frequency filtering module that combines adaptive average pooling and differentiable filtering to effectively extract texture details that are difficult to capture with conventional Fourier transforms. In the residual path, a learnable scaling factor was introduced to dynamically adjust the fusion ratio between residual and identity mappings, thereby improving the flexibility of feature representation and enhancing training stability. Subsequently, the Lightweight Transformer Block (LTB), which integrates Efficient Transformer (ET) units, was applied. The LTB employs a local window partition strategy to replace global attention, thus reducing computational cost, while jointly modeling long-range dependencies in images through multi-head attention and feed-forward networks. Finally, image reconstruction was performed through a hybrid upsampling module, which combines pixel-shuffle and element-fusion operations, achieving spatial resolution enhancement and multi-scale information integration.

3.1.2. SR- ESRT architecture

To improve SR reconstruction performance under small-sample conditions, this study proposes a joint training strategy that combined conventional optical image datasets with a limited set of THz-TDS point-scan images. Based on this strategy, an SR-DESRT module was designed, as illustrated in Fig. 3. The SR-DESRT consists of two independent ESRT networks: (i) ESRT-DIV2K, which was pre-trained on the publicly available DIV2K grayscale dataset and was kept frozen during the training stage. This design helped alleviate the limited generalization capacity caused by the small number of THz-TDS training samples; (ii) ESRT-THz, which was fine-tuned using a small batch of THz-TDS point-scan images, thereby enhancing the model's ability to capture frequency-domain features specific to THz-TDS imaging. The feature maps generated by the two networks were concatenated along the channel dimension to form a unified representation F, which was then passed to the subsequent channel fusion module.

In this SR reconstruction process, images from different frequency points were processed independently at the network input level, rather than being stacked and jointly processed as multi-frequency inputs in a single forward pass. Specifically, images of the same sample at multiple frequency points were fed into the network separately, while multiple single-frequency images were processed simultaneously in each batch. This strategy not only reduced the computational burden during training and inference but also made the framework more suitable for efficient

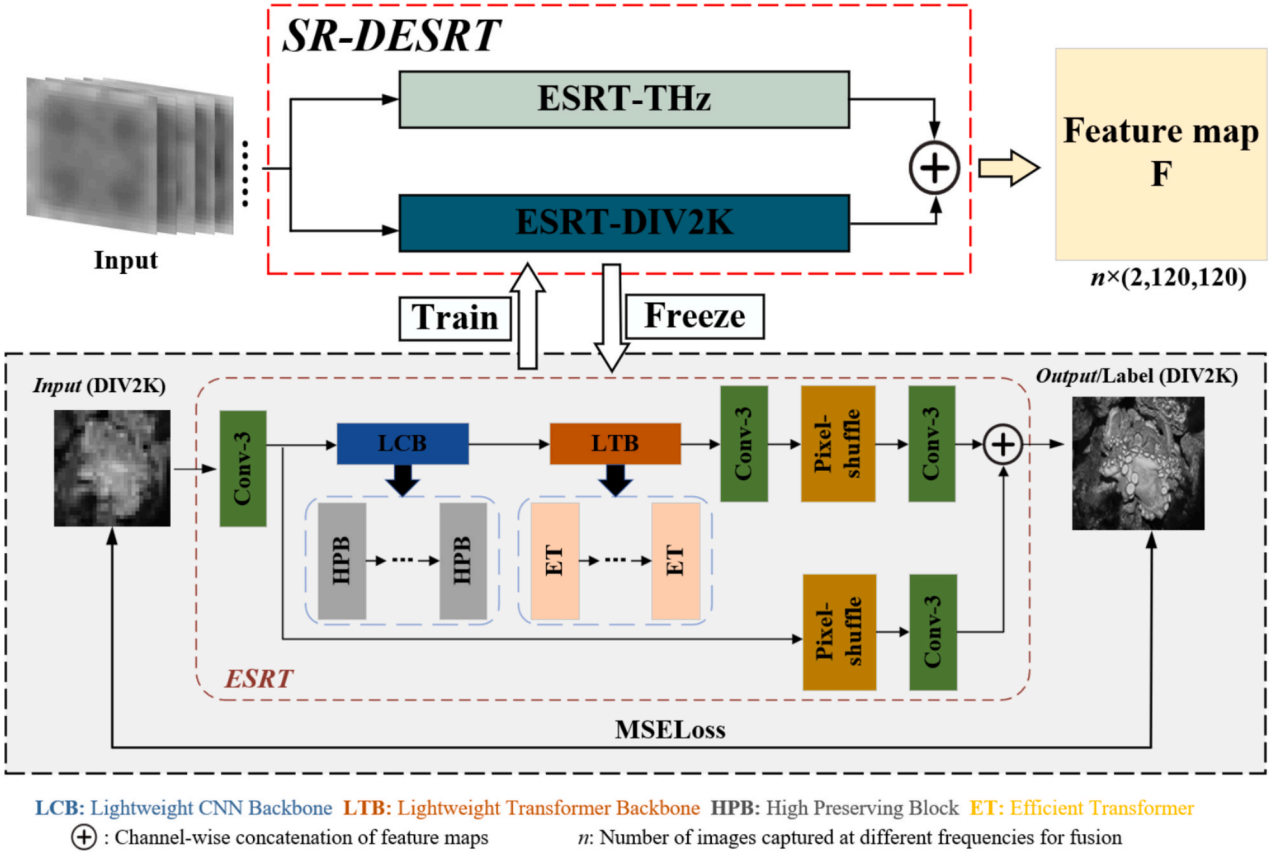


Fig. 3. Network structures of SR-DESRT.

network deployment. Moreover, it helped alleviate the risk of overfitting under data-limited conditions.

3.1.3. Channel adjustment based on ResBlockCBAM

Since the two ESRT branches differed in training strategies and data domains, their output features exhibited significant heterogeneity. To achieve effective integration of these features and enhance both adaptability and discriminability in the fusion representation, we proposed a channel adjustment module based on a joint attention mechanism—CA-ResBlockCBAM, as illustrated in Fig. 4. This module consists of multiple residual blocks embedded with the CBAM attention mechanism (ResBlockCBAM) [32,33] and 2D convolutional layers, which enable adaptive feature reweighting along both channel and spatial dimensions. In particular, the Channel Attention Mechanism (CAM) extracts global channel information via max pooling and average pooling, which is then passed through a fully connected network followed by a Sigmoid activation to generate channel weights M_c. This operation strengthens the relative importance of optical and THz-TDS image channels during feature fusion [34], as formulated in Eq. (1). The Spatial Attention Mechanism (SAM), on the other hand, emphasizes critical spatial regions of the image. It applies average pooling and max pooling operations on the feature maps, concatenates them along the channel dimension, and processes them through a convolution layer with Sigmoid activation to generate spatial attention weights M_s, as formulated in Eq. (2). This improves reconstruction performance at different spatial locations and improves the overall image restoration quality. Furthermore, the residual connections incorporated in this module facilitate gradient propagation during training, thereby improving network stability and convergence efficiency. By applying deep feature extraction and channel adjustment to the concatenated multi-source feature map F, the CA-ResBlockCBAM produces the output for the first-stage SR reconstruction.

$$\boldsymbol{M}_c = \sigma\left(\mathrm{MLP}(\mathrm{AvgPool}(X)) + \mathrm{MLP}(\mathrm{MaxPool}(X))\right) \tag{1}$$

$$\boldsymbol{M}_s = \sigma\left(\mathrm{Conv}(\mathrm{AvgPool}(X);\ \mathrm{MaxPool}(X))\right) \tag{2}$$

where σ denotes the Sigmoid activation function; MLP represents the fully connected mapping; AvgPool denotes average pooling; X is the feature map; MaxPool denotes max pooling; and Conv represents the convolution mapping.

It is important to note that during the entire first-stage training process, the weights of the ESRT-DIV2K branch were kept frozen, while only the parameters of the ESRT-THz and CA-ResBlockCBAM modules were updated. Training was fully driven by the small-batch THz-TDS point-scan images. This design not only preserved the prior advantage of optical images in spatial detail modeling but also enabled deep integration of features through the adaptive channel fusion mechanism, thereby significantly enhancing the reconstruction capability and generalization performance of the model under data-scarce conditions.

3.2. Multi-frequency image fusion based on genetic algorithm

Although the first-stage SR reconstruction effectively improves the spatial detail representation of single-frequency images, the quality and information content still vary across different frequencies. Relying solely on manually selecting the optimal frequency image as the basis for subsequent processing does not meet the automation and robustness requirements of practical nondestructive testing. Moreover, THz-TDS images at different frequencies often contain complementary information, and the rational fusion of such information can further improve structural integrity and the representation of critical features in the final reconstructed image. To this end, we propose a MFIF-GA method as the second-stage SR reconstruction module of the DL-MFIF framework. This approach adaptively fuses multiple frequency-point images generated in the first stage by optimizing fusion weights and adaptively adjusting image parameters, thereby achieving automatic high-quality image reconstruction. GA is a typical heuristic global optimization method characterized by high search efficiency, strong adaptability, and the ability to escape local optima [35]. These advantages make it particularly suitable for image processing tasks in which prior knowledge is unavailable and the objective function is highly complex. The overall process is illustrated in Fig. 5(a). Unlike the classical GA, the GA employed in this study incorporates an elitism strategy, where the top m % of individuals with the highest fitness scores in the current population are directly preserved in the next generation. This design prevents crossover and mutation operations from destroying the chromosomes of the best-performing individuals.

Fig. 5(b) illustrates the population initialization mechanism of MFIF-GA. In this population, each individual is encoded with three types of parameters—Gaussian denoising kernel parameters, contrast adjustment coefficients, and image fusion weights—corresponding to the feature processing of images at different frequency points. Since all three types of parameters are floating-point values within the same ranges, continuous probability distributions were employed for their initialization. Specifically, the fusion weights were initialized using a Dirichlet distribution [36], as expressed in Eq. (3), while the Gaussian denoising parameters and contrast adjustment coefficients were initialized using a uniform distribution.

The fusion process is illustrated in Fig. 5(c). First, each single-frequency SR image obtained from the first stage was processed with Gaussian denoising and contrast enhancement. Gaussian denoising aimed to further remove residual noise in the images, thereby reducing the noise level in the second-stage SR fusion results, as formulated in Eq.

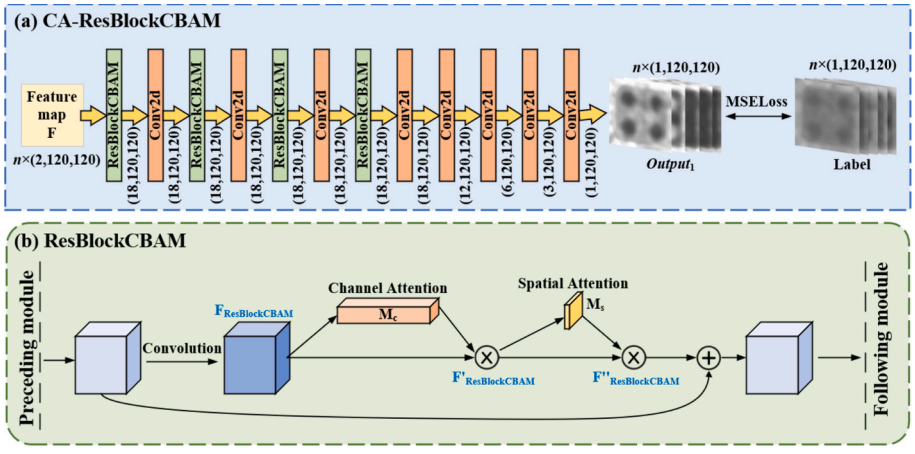


Fig. 4. Network structures of CA-ResBlockCBAM and ResBlockCBAM.

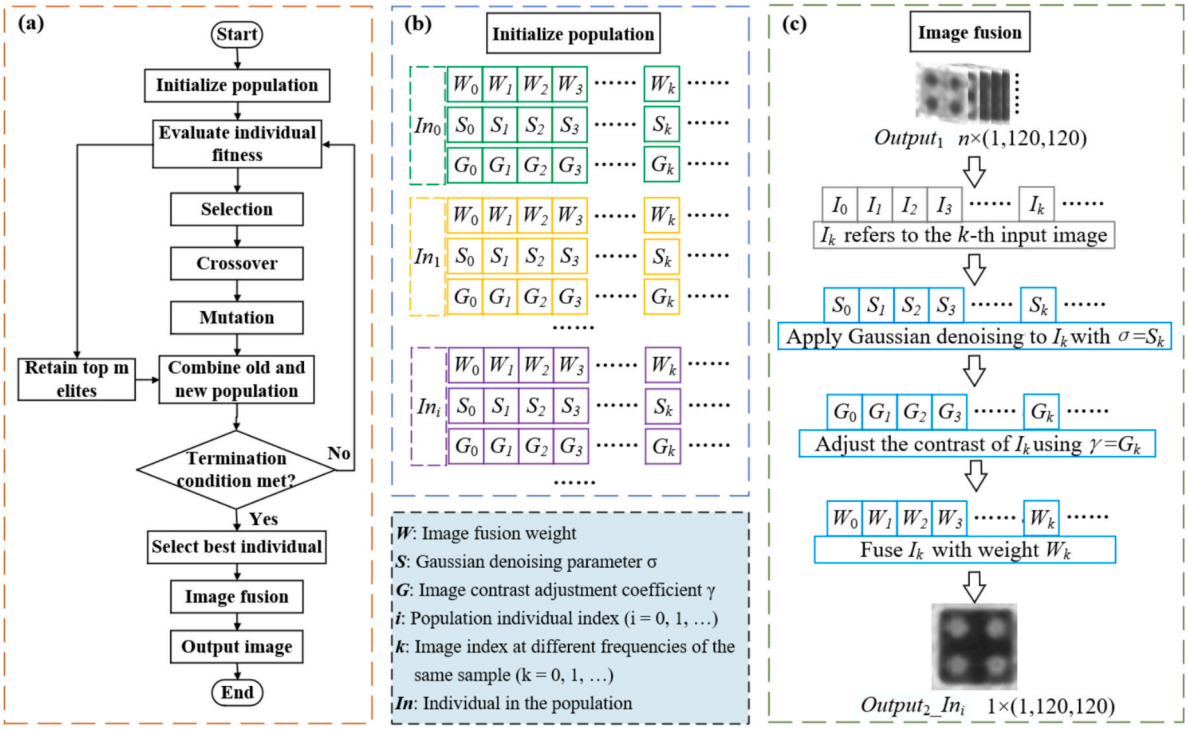


Fig. 5. MFIF-GA (a) overall workflow; (b) population initialization; (c) image fusion.

(4). Contrast adjustment was applied to enhance critical features in the fused image and improve the visibility of key structures in the second-stage SR output, as expressed in Eq. (5). Finally, a weighted averaging operation was performed based on the fusion weights to align pixels across images from different frequency points, yielding a single fused image as the final result, as formulated in Eq. (6).

$$Dir(\mu|\alpha) = \frac{\Gamma(\alpha_0)}{\prod_{k=1}^{K}\Gamma(\alpha_k)} \prod_{k=1}^{K} \mu_k^{\alpha_k - 1} \tag{3}$$

$$\text{Gauss}(i,j) = \frac{1}{2\pi\sigma^2} e^{-\frac{i^2+j^2}{2\sigma^2}} \tag{4}$$

$$\text{Enhancement}(i,j) = \gamma(I(i,j) - 0.5) + 0.5 \tag{5}$$

$$I_{\text{out}} = \sum_{k=0}^{k=n} W_k I_k' \tag{6}$$

where μ denotes the image fusion weight value; $\alpha=(\alpha_1, \alpha_2, \ldots, \alpha_K)$ represents a vector of positive real parameters, $\alpha_0 = \sum_{k=1}^{K}\alpha_k$; $\Gamma(\cdot)$ is the Gamma function; K is the total number of variables, which in this study corresponds to the total number of frequency-point images n; k is the index of images from the same sample at different frequency points; x represents the Gaussian denoising or contrast enhancement parameter value of a given frequency-point image; i and j denote the horizontal and vertical pixel indices of the image, respectively; σ denotes the kernel parameter; W is the image fusion weight; I' represents the image obtained after Gaussian denoising and contrast adjustment. For the Dirichlet distribution, the parameters satisfy $\mu_k \in [0, 1]$ and $\sum_{k=1}^{K}\mu_k = 1$. The pixel intensity values were normalized to the range [1]. Since $W_k \sim \text{Dir}(\mu|\alpha)$, it follows that $\sum_{k=1}^{n} W_k = 1$.

To ensure that the fusion results achieved both edge sharpness and low noise levels, this study employed a weighted combination of the Sobel gradient function (SMD) and the no-reference signal-to-noise ratio (NRI-SNR) as the fitness function. The SMD captured effective detail features through its sensitivity to local gradients and served as a measure of image detail clarity. Its theoretical foundation stemmed from the spatial-frequency mapping characteristics of optical systems: under ideal focusing conditions, the image exhibited maximized high-frequency energy, which in the spatial domain corresponded to the maximization of the cumulative absolute gray-level differences between adjacent pixels. The computation of SMD is given in Eq. (7). The NRI-SNR approximated the separation of signal and noise by establishing a statistical model. Specifically, the mean pixel intensity reflected the spatial distribution of signal strength (the mathematical expectation of an image is typically robust to random noise), whereas the variance characterized the degree of pixel dispersion (the image variance is generally positively correlated with noise energy). By leveraging global statistical properties, the NRI-SNR reduced the influence of pseudo-gradient responses caused by high-frequency noise and served as an indicator of the relative signal-to-noise level of the image. Its computation is defined in Eq. (8). Finally, the two metrics were linearly combined for adaptive fusion, as shown in Eq. (9), where both α and β were set to 0.5, assigning equal importance to detail enhancement and noise robustness. This choice reflected the need to balance edge preservation and noise suppression in THz imaging, as excessive emphasis on gradient-based measures might amplify noise, whereas excessive emphasis on noise suppression might over-smooth structural details. Based on this setting, several weight combinations in the vicinity of $\alpha = \beta = 0.5$ were evaluated, and the fusion results remained stable within a reasonable range of weight variations. Therefore, to avoid introducing additional tuning complexity, fixed equal weights were adopted in all experiments.

$$\text{SMD}(I) = \sum_{j}\sum_{i}(|I(i,j) - I(i,j-1)| + |I(i,j) - I(i+1,j)|) \tag{7}$$

$$\text{NRI - SNR}(I) = \frac{\text{mean}(I)}{\sqrt{\text{var}(I)}} \tag{8}$$

$$\text{Fitness}(x) = \alpha\text{SMD}(x) + \beta\text{NRI} - SNR(x) \tag{9}$$

where i and j represent the horizontal and vertical pixel indices, respectively; *var* denotes the variance of all pixel values; and *mean*

represents the average intensity of all pixel values; I denotes the image used for fitness evaluation; x denotes the processing result of individual in the population; α and β are the weights of SMD and NRI-SNR, respectively.

Through multi-generation evolutionary iterations, MFIF-GA adaptively determined the fusion parameters for each frequency-point image of the specified sample and generated a single high-quality fused image. This fused image served as the output of the second-stage SR reconstruction.

4. Experiments and results analysis

4.1. Dataset construction

4.1.1. THz-TDS system and parameter settings

In this study, the TAS7400TS THz-TDS system (Advantest, Japan) was utilized, as shown in Fig. 6. The system mainly consists of a detection optical path, a control computer, a THz laser source, an analysis unit, a motion control unit, and a drying unit. The optical path measurement setup primarily includes a THz emitter, an x–y motorized translation stage, and a THz receiver. The THz emitter generates broadband THz waves by exciting a LiNbO3 crystal waveguide with femtosecond pulsed laser irradiation. The x–y translation stage enables stepwise scanning of the sample via motor control. The THz receiver detects THz wave signals using a photoconductive switch triggered by femtosecond pulsed laser excitation. The femtosecond laser used in the system operates at a wavelength of 1550 nm, with a repetition rate of 50 MHz and a pulse width of less than 50 fs. The system provides a dynamic range of up to 60 dB, a maximum delay range of 131 ps, and an effective spectral range of 0.1–4 THz. It supports both transmission and reflection measurements as well as THz-TDS point-scan imaging. The scanning speed is 200 ms per measurement point, with spectral resolution options of 1.9 GHz and 7.6 GHz. The imaging resolution reaches up to 200 μm/pixel, with spectral accuracy better than ± 10 GHz, and the maximum imaging size is 150 mm × 150 mm. In addition, the system ensures that the peak error of at least five repeated measurements at the same point remains within 1%.

4.1.2. Dataset

Two data sources were used for model pre-training and fine-tuning: the publicly available single-image SR dataset DIV2K and a self-collected THz-TDS point-scan dataset of photosensitive resin samples. The DIV2K dataset consists of high-quality RGB natural images and has been widely used in single-image SR tasks [37]. In this study, it was utilized for pre-training the ESRT-DIV2K branch. The THz-TDS dataset, acquired using the TAS7400TS THz-TDS system, was mainly used for fine-tuning the ESRT-THz branch, training the CA-ResBlockCBAM module, and evaluating the final reconstruction performance of the proposed model.

For the DIV2K dataset, additional preprocessing was performed to improve the transferability of the pre-trained module to the THz-TDS image reconstruction task. First, all DIV2K images were converted into grayscale. Then, paired high- and low-resolution images were constructed: the high-resolution images were downsampled via bilinear interpolation to 120 × 120 pixels, while the low-resolution images were downsampled to 30 × 30 pixels. To simulate the spatial degradation characteristics of THz-TDS imaging, anisotropic Gaussian blurring was applied to approximate the PSF. Specifically, the downsampled 30 × 30 pixel DIV2K images were convolved with Gaussian kernels [18], as formulated in Eq. (10). To enhance data diversity and simulate varying levels of image degradation, the kernel size was randomly selected from the set {3 × 3, 5 × 5, 7 × 7, 9 × 9, 11 × 11}. In total, 800 pairs of low–high resolution DIV2K image samples were constructed and used for ESRT-DIV2K pre-training.

$$g(x, y, \sigma_x, \sigma_y) = \frac{1}{\sqrt{2\pi\sigma_x\sigma_y}} e^{-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2}+\frac{y^2}{\sigma_y^2}\right)} \tag{10}$$

where x and y denote the horizontal and vertical pixel coordinates, respectively; σ_x and σ_y represent the Gaussian blur variance factors in the horizontal and vertical directions, with the constraints $\sigma_x \in [1, 9]$ and $\sigma_y \in [1, 9]$.

The samples for the THz-TDS point-scan image dataset were fabricated using 9400R photosensitive resin via high-precision 3D printing (accuracy: 0.1 mm). A total of 10 samples with different structural patterns were prepared, each with dimensions of 40 mm × 40 mm. The front and back views of the samples are shown in Fig. 7(a) and Fig. 7(b). THz-TDS images were acquired using the transmission scanning mode, and high-quality point-scan data were extracted using the filter peak attention network (FPAN) method proposed by Zhang et al. [38]. Two scanning strategies were applied: a small-step scan of 120 × 120 pixels with a step size of 0.4 mm, and a large-step scan of 30 × 30 pixels with a step size of 1.5 mm. The spectral resolution was set to 7.6 GHz, and for each sample, 12 frequency-domain images were collected over the range of 0.1–1.2 THz at 0.1 THz intervals, in addition to one time-domain maximum image. Thus, 13 paired images were obtained for each sample. In total, 130 pairs of THz-TDS point-scan images were collected and further augmented by rotation and mirroring to yield 1690 pairs. An example of image pairs at 0.1 THz is presented in Fig. 7(c), where the left image of each pair corresponds to large-step scanning and the right image to small-step scanning. It can be observed that even the small-step images contain noticeable point-scan noise and background artifacts, indicating that the high-resolution reference images are themselves non-ideal. This observation further underscores the necessity of improving SR reconstruction quality for THz-TDS images. It should be noted that although data augmentation increased the dataset size by a factor of 13,

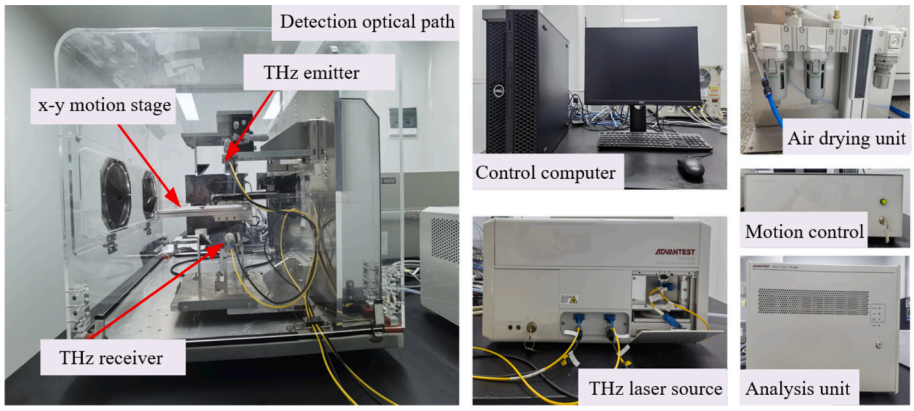
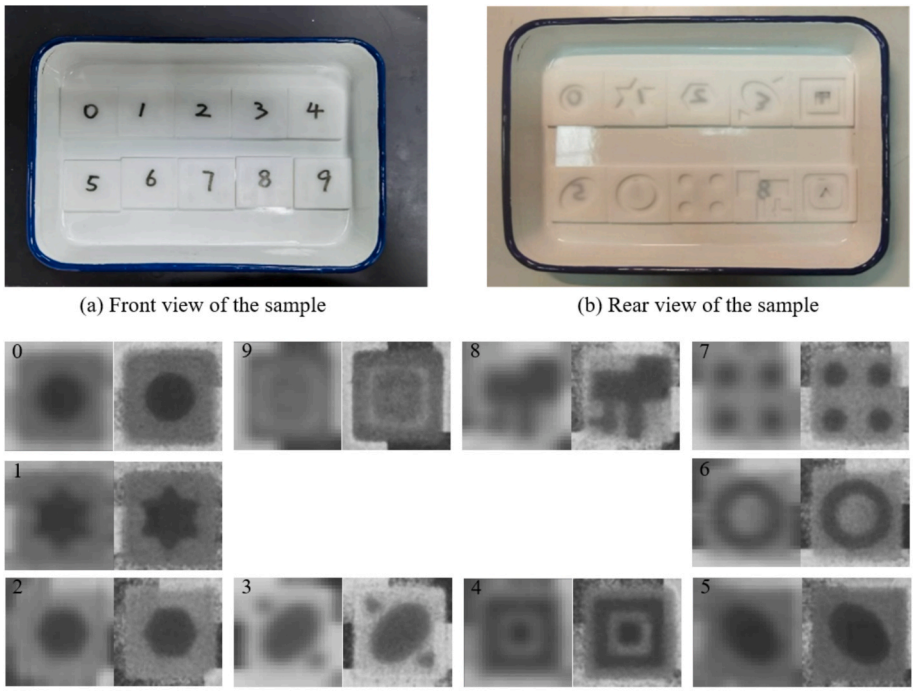


Fig. 6. TAS7400TS THz-TDS system.



(a) Front view of the sample

(b) Rear view of the sample

(c) Paired THz imaging data (0.1 THz): For each sample numbered 0–9, the left image shows the result from large-step scanning and the right image from small-step scanning

Fig. 7. THz-TDS image dataset.

the dataset essentially remains small-scale, with morphological similarities among samples that may lead to overfitting during model training. Therefore, the natural image dataset DIV2K was incorporated for pre-training, which not only enriched the spatial texture priors of the model but also significantly enhanced its generalization ability and stability in THz-TDS image reconstruction tasks.

The THz-TDS image dataset was divided into training and testing sets at a ratio of 8:2, where samples No. 6 and No. 7 were designated as the testing set, and the remaining samples were used for model training. Correspondingly, the DIV2K dataset used for ESRT-DIV2K pre-training was also partitioned using the same 8:2 strategy.

4.2. Comparative experimental design and model configuration

To comprehensively evaluate the effectiveness and technical contribution of the proposed DL-MFIF framework for small-sample THz image super-resolution reconstruction, both external method comparisons and internal ablation studies were conducted.

For external comparisons, bilinear interpolation was first adopted as a conventional upsampling baseline to assess the performance gains of deep learning–based methods over traditional numerical interpolation techniques. Subsequently, to evaluate the performance of the proposed approach over single deep convolutional networks, comparisons were performed against the hybrid transformer super-resolution GAN (HTSRGAN) [17] and the residual channel attention network (RCAN) [18]. RCAN represents a typical CNN-based architecture widely used in THz imaging. Its core structure consists of a shallow feature extraction module, deep feature mapping composed of six residual channel attention groups (RCAGs), and an image reconstruction module. Each RCAG integrates residual blocks with squeeze-and-excitation (SE) channel attention, enabling adaptive recalibration of channel-wise feature responses to enhance the representation of high-frequency texture information. HTSRGAN belongs to the class of generative models and combines the capability of Transformers to model long-range dependencies with the texture synthesis ability of GAN discriminators. Its generator comprises six hybrid Transformer modules with four-head self-attention mechanisms to capture global spatial dependencies, while the discriminator adopts a VGG-based deep convolutional architecture to distinguish generated images from real ones. During training, a composite loss function was employed, including a pixel-wise loss (weight 10), a VGG19-based perceptual loss (weight 1), and an adversarial loss (weight 0.005).

For ablation studies, to verify the effectiveness of each component in DL-MFIF, the image quality performance of ESRT-DIV2K, ESRT-THz, SR-DESRT + CA-ResBlockCBAM, and DL-MFIF was compared on the test set. Specifically, ESRT-DIV2K was used to evaluate the adaptability of transfer learning strategies to THz images; ESRT-THz assessed the performance achievable when training solely on a limited amount of THz-TDS data; and SR-DESRT + CA-ResBlockCBAM represented the complete first-stage deep learning architecture, enabling analysis of the contribution of the multi-frequency fusion module and validation of the rationality of the two-stage design.

In addition, to evaluate the role of the GA in optimizing multi-frequency fusion parameters, two comparative fusion strategies were introduced: gradient-based fusion (GF) and metric-based weighted averaging (MWA). The GF method computed gradient magnitude maps for each single-frequency image using the Sobel operator and reconstructed the final image by selecting, at each pixel location, the pixel with the maximum gradient magnitude, thereby selecting the most prominent features across frequencies. The MWA strategy assigned global scores to each single-frequency image based on the entropy quality metric, normalized these scores, and applied the resulting coefficients to perform linear weighted aggregation. By comparing image quality metrics under different fusion strategies, the experiments verified that MFIF-GA was capable of automatically identifying suitable fusion parameters according to the image-level measurement characteristics, thereby more effectively exploiting the complementary information among multi-frequency THz data.

In this study, the mean squared error (MSE) loss was adopted as the training loss function for all deep learning models. For the experimental parameter settings, the batch size was set to 16, the number of training epochs was set to 200, and the initial learning rate was set to 1×10^{-4},

with a minimum learning rate of 1×10^{-6}. A cosine annealing scheduler was applied to optimize the learning rate, and the Adam optimizer was employed. For the GA, the initial population size was set to 6000, the number of evolutionary iterations was set to 20, the mutation probability was set to 25%, the elitism rate was set to 10%, and the maximum population size was limited to 7000.

Regarding the hardware environment, the training and inference of the deep learning-based SR reconstruction models were conducted on a workstation equipped with an Intel Xeon Silver 4314 CPU @ 2.40 GHz and dual NVIDIA GeForce RTX 4090 GPUs, while the MFIF-GA computation process was mainly executed on the CPU.

4.3. Evaluation metrics

To evaluate the effectiveness of the proposed method for THz image super-resolution reconstruction, both no-reference and reference-based image quality metrics were employed for quantitative assessment. The no-reference metrics included variance, contrast, information entropy, and average gradient, while the reference-based metrics comprised peak signal-to-noise ratio (PSNR), structural similarity index (SSIM), and learned perceptual image patch similarity (LPIPS). It should be emphasized that, in this study, the high-resolution images used for training and evaluation are reference measurements acquired via small-step point-scan scanning.

Among these, variance, contrast, and information entropy serve as global measures based on spatial distribution, while average gradient serve as a structural detail metric based on image gradients. Variance quantifies the dispersion of pixel values relative to the mean of the entire image. For a clear image, the pixel intensity distribution is wider, resulting in a higher variance. Contrast reflects the distribution of brightness levels in an image and is essentially the ratio of the maximum to minimum intensity. Higher contrast provides richer information and more detailed visual representation. To overcome the numerical dilution issue of traditional global contrast measures (e.g., Michelson contrast), particularly in high-brightness regions, we employed a local contrast metric based on a non-overlapping sliding window. Specifically, the global image was divided into multiple local sub-images using an 8×8 window with a stride of 8. The maximum–minimum gray-level difference was calculated for each sub-image, and their mean value was defined as the overall image contrast. This approach better captures detail-level contrast and avoids the dilution effect when the denominator greatly exceeds the numerator in bright regions. Information entropy represents the average information level of an image. Higher entropy indicates a more uniform gray-level distribution and richer detail content. Average gradient characterizes the degree of local detail variation in an image. It was computed using Sobel operators to extract horizontal and vertical gradients, from which the mean gradient was derived. For a clear image, variations are more pronounced, resulting in a higher average gradient. PSNR measures the discrepancy between the reconstructed image and the reference image in terms of pixel-wise error. It is essentially a logarithmic metric derived from the mean squared error (MSE). A higher PSNR value indicates that the reconstructed image is closer to the reference image at the pixel level. SSIM evaluates the similarity between two images by jointly considering luminance, contrast, and structural information, and is therefore more consistent with human visual perception. The SSIM value ranges from 0 to 1, with higher values indicating greater structural and perceptual similarity between the two images. LPIPS is a perceptual similarity metric based on deep neural network features, which quantifies visual similarity by comparing differences between images in a high-level semantic feature space. Unlike PSNR and SSIM, LPIPS places greater emphasis on the overall perceptual consistency of textures, structures, and semantic content as perceived by the human visual system. A lower LPIPS value indicates higher perceptual similarity between the reconstructed and reference images.

The testing set (samples No. 6 and No. 7) was chosen as the evaluation target. It is worth noting that different strategies were adopted for selecting evaluation images across the compared methods. For the proposed DL-MFIF method, the evaluation metrics were calculated based on the final SR image generated for each sample. For single-stage methods without the MFIF-GA module, the evaluation followed the fusion weight results obtained by MFIF-GA, where the frequency-point image with the highest fusion weight was selected as the representative evaluation image. Consequently, each method corresponded to two evaluation images (samples No. 6 and No. 7), and the average of the results from these two images was reported as the final metric value.

4.4. Quantitative comparison of experimental results

Table 1 presents the quantitative image quality evaluation results of different methods on the test set. The adopted metrics can be grouped into two categories. Variance, contrast, information entropy, and average gradient mainly characterize the overall information content, contrast distribution, and structural detail strength of the reconstructed images. In contrast, PSNR, SSIM, and LPIPS assess reconstruction performance with respect to the high-resolution reference image acquired using small-step scanning as a reference measurement, from the perspectives of pixel fidelity, structural similarity, and perceptual similarity, respectively.

As shown in the results, conventional bilinear interpolation exhibited the weakest performance across all metrics. Its variance, contrast, average gradient, and information entropy remained at the lowest levels, while its PSNR, SSIM, and LPIPS indicated a large discrepancy from the reference image, highlighting the inherent limitations of traditional interpolation methods in THz super-resolution tasks. The ablation results further revealed that ESRT-DIV2K leveraged spatial priors learned from natural images; however, due to the pronounced domain shift, its PSNR reached only 20.8 dB. In contrast, ESRT-THz was constrained by the limited amount of training data and was unable to sufficiently recover structural details. Its variance (1459.3) and average gradient (31.4) were lower than those of ESRT-DIV2K, indicating that learning effective high-frequency texture mappings from scarce THz data alone was challenging.

After introducing the channel adjustment module, the SR-DESRT + CA-ResBlockCBAM model achieved a clear improvement across multiple metrics. Its variance, contrast, and average gradient were substantially higher than those of the single-branch ESRT variants, and the PSNR increased to 21.0 dB. These results demonstrated that the multi-branch architecture combined with channel attention enhanced feature representation capability, alleviated the small-sample limitation, and improved reconstruction quality.

The comparison among different fusion strategies further verified the necessity of the proposed MFIF-GA scheme. The GF method achieved high sharpness-related metrics, with an average gradient of 48.5 and a contrast of 49.8. However, its PSNR, SSIM, and LPIPS did not reach optimal levels simultaneously, indicating that excessive emphasis on gradient information tended to amplify noise and degrade signal-to-noise ratio and structural consistency. The MWA strategy achieved a more balanced performance across several metrics, with PSNR increasing to 21.7 dB, SSIM reaching 0.845, and LPIPS decreasing to 0.145, suggesting cleaner overall reconstructions. Nevertheless, its variance (2350.2) and average gradient (41.5) were notably lower than those of the GF method, implying a loss of edge sharpness and blurred structural boundaries. By contrast, DL-MFIF employed a genetic algorithm to search for suitable fusion parameters in the global solution space, enabling the final model to achieve the most balanced performance across all evaluated metrics. Specifically, PSNR increased to 22.1 dB, SSIM reached 0.889, and the model maintained the highest variance (3147.8) and average gradient (49.1). These results indicated that DL-MFIF effectively preserved high-frequency details comparable to GF while simultaneously achieving the noise suppression and structural consistency characteristic of MWA.

Table 1
Evaluation metrics of each method on the test set.

Method	Variance	Contrast	Information Entropy	Average Grad	PSNR	SSIM	LPIPS
ESRT-DIV2K	1537.5	35.9	7.0	34.2	20.8	0.749	0.196
ESRT-THz	1459.3	33.4	7.0	31.4	20.7	0.760	0.185
SR-DESRT + CA-ResBlockCBAM	1985.2	42.4	7.1	39.7	21.0	0.752	0.160
SR-DESRT + CA-ResBlockCBAM + GF	2950.5	49.8	7.25	48.5	21.1	0.815	0.162
SR-DESRT + CA-ResBlockCBAM + MWA	2350.2	45.6	7.18	41.5	21.7	0.845	0.145
Bilinear interpolation	1361.1	31.0	6.94	28.7	20.5	0.772	0.232
RCAN	2584.3	50.3	7.3	48.2	21.2	0.857	0.155
HTSRGAN	2790.7	50.7	7.2	48.0	20.6	0.841	0.148
DL-MFIF	**3147.8**	**52.5**	**7.4**	**49.1**	**22.1**	**0.889**	**0.136**

Among the external comparison methods, RCAN and HTSRGAN achieved relatively high no-reference metric values, particularly in terms of variance, contrast, and average gradient. However, RCAN was constrained by the smoothing tendency of the L1 loss, and its LPIPS (0.155) and variance (2584.3) remained lower than those of the proposed method, revealing limitations in recovering fine-scale textures. HTSRGAN achieved high contrast (50.7), average gradient (48.0), and a favorable LPIPS value (0.148). Nevertheless, GAN-based methods typically traded pixel fidelity (PSNR) for perceptual quality (SSIM and LPIPS). Consequently, its PSNR was only 20.3 dB, and its SSIM (0.841) was also lower than that of RCAN. This observation suggested that under small-sample conditions, large-scale Transformer-GAN architectures showed difficulty in achieving optimal fidelity.

In comparison, DL-MFIF achieved the highest structural fidelity while maintaining superior perceptual quality, with the lowest LPIPS value of 0.136 and the highest PSNR of 22.1 dB. Specifically, DL-MFIF improved PSNR by 0.9 dB over RCAN and increased SSIM by 0.048 compared with HTSRGAN. Given the inherently high noise level and alignment uncertainty in THz imaging, substantial improvements in PSNR were difficult to obtain. Therefore, the more meaningful performance gains were reflected in enhanced structural perception and detail recovery, as evidenced by improvements in SSIM, LPIPS, and gradient-based metrics.

4.5. Qualitative comparison of experimental results

As shown in Fig. 8, images at different frequency points exhibited varying levels of feature clarity and detail representation. Low-frequency images displayed the clearest structural features but inevitably suffered from a certain degree of detail loss. The proposed DL-MFIF method significantly improved image details, edge transitions, and overall visual quality. This improvement was mainly attributed to the MFIF-GA module, which tended to assign higher fusion weights to low-frequency images while simultaneously learning optimal image enhancement parameters. As a result, the final SR reconstruction achieved notable gains in sharpness and detail representation. As shown in Fig. 8, samples 6 and 7 showed the highest fusion weights at 0.1 THz. Therefore, the images at this frequency were selected as the evaluation images for the comparative methods and were further compared with the fusion results obtained by the proposed DL-MFIF method in terms of detail representation.

Fig. 9 presents a comparison of detail reconstruction in SR results obtained by different methods. The bilinear interpolation results appeared the most blurred and lacked fine structural details. The ESRT-DIV2K results exhibited mosaic-like noise, as the model failed to adequately learn the specific mapping characteristics of THz-TDS point-scan images during training, which limited its generalization to THz-TDS data. Although ESRT-THz demonstrated a certain level of SR reconstruction capability, it was still insufficient in restoring fine structural details. In contrast, SR-DESRT + CA-ResBlockCBAM achieved significant improvements in sharpness and texture feature. RCAN performs better than interpolation-based methods in edge restoration and is able to reconstruct the circular ring contours of Sample 6 reasonably well. However, in local regions of Sample 7, noticeable granular noise is observed. In these areas, high-frequency noise is mistakenly enhanced as texture details, resulting in a visually rough appearance. The results produced by HTSRGAN exhibit sharper edges and textures, but also introduce a considerable number of artificial details and noise. These artifacts, induced by adversarial training, lead to improved LPIPS values, yet lead to texture patterns that are inconsistent with the true underlying structures at the pixel level, which is consistent with the relatively low PSNR values reported in Table 1. In contrast, the proposed DL-MFIF method achieves a more balanced trade-off between detail enhancement and noise suppression. For Sample 6, the reconstructed ring structures show clear intensity separation and well-defined edges among all methods. For Sample 7, the four circular features are more distinctly separated, with minimal artifacts and noise, indicating improved reconstruction fidelity compared with the other methods.

Table 2 presents the time cost comparison between the proposed method and the actual THz point-scanning data acquisition process for single-sample imaging. As shown in the table, when only the first-stage deep learning–based super-resolution reconstruction module (SR-DESRT + CA-ResBlockCBAM) was employed, the inference time for a single image was approximately 0.65 s, indicating high computational efficiency. After introducing the MFIF-GA module, the overall runtime increased noticeably. This increase was mainly attributed to the global optimization of multi-frequency fusion parameters required by the GA, which introduced additional computational overhead. Nevertheless, the DL-MFIF framework avoided manual selection of frequency points with relatively good imaging quality and enabled adaptive multi-frequency fusion, thereby further improving the reconstructed image quality.

Although the total runtime of DL-MFIF reached about 6.5 min, compared with using high-density scanning with a small step size to achieve comparable spatial detail, it still reduces the overall imaging time by more than 80%, demonstrating its practical feasibility for engineering applications. For applications with high real-time requirements, the first-stage single-image super-resolution reconstruction method (SR-DESRT + CA-ResBlockCBAM) exhibited both high efficiency and stable performance, and could therefore serve as a fast imaging solution.

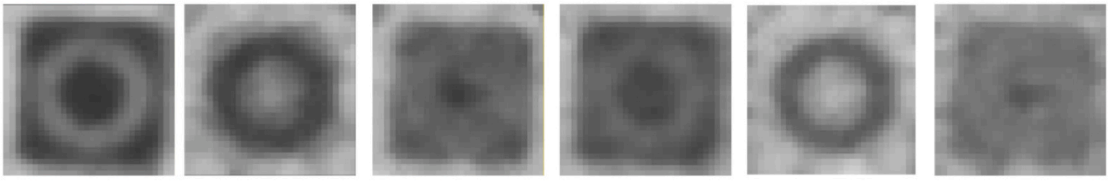
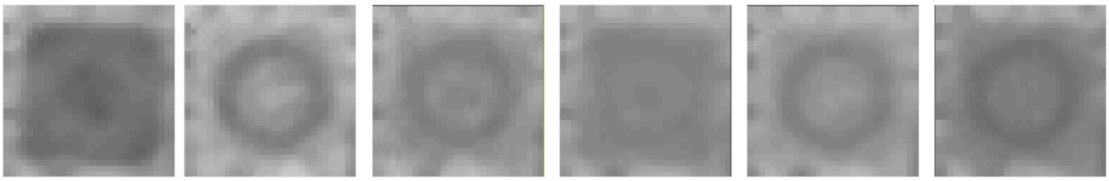
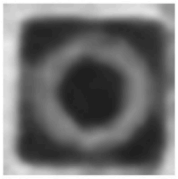
Overall, the proposed SR framework shows consistent improvements in both quantitative evaluation metrics and qualitative visual results, indicating enhanced adaptability, reconstruction quality, and measurement efficiency in THz-TDS point-scanning imaging for nondestructive testing.

4.6. Discussion

First, regarding the dataset scale and model generalization, this study employed 10 photosensitive resin specimens, yielding a total of 130 original image pairs. Compared with large-scale computer vision datasets, this sample size was indeed limited. This limitation was primarily due to the extremely high time cost of THz point-scanning imaging, where a single high-resolution scan required approximately 48 min, making the acquisition of large-scale paired high-/low-resolution data highly challenging from an engineering perspective. Within this

(a)

Sample 6

Frequency /THz	0.1	0.2	0.3	0.4	0.5	0.6
Low-resolution original image						
Fusion weight	9.38e-1	2.73e-4	2.73e-4	2.73e-4	2.73e-4	9.43e-3
σ	1.13	5.77	1.38	6.63	6.80	2.02
γ	1.30	1.17	1.14	0.94	1.3	0.96
Frequency /THz	0.7	0.8	0.9	1.0	1.1	1.2
Low-resolution original image						
Fusion weight	9.43e-3	9.43e-3	9.43e-3	9.43e-3	9.43e-3	4.74e-3
σ	1.04	0.83	4.96	6.14	1.47	6.77
γ	0.87	1.02	0.56	0.77	0.82	0.78
High-resolution result image and its 3D sample structure						

(b)

Sample 7

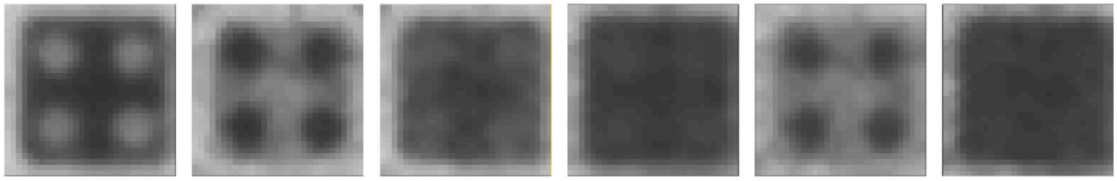
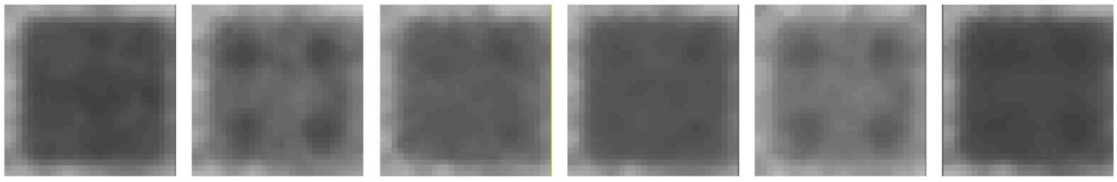
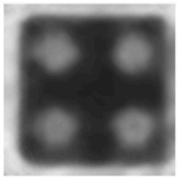
Frequency /THz	0.1	0.2	0.3	0.4	0.5	0.6
Low-resolution original image						
Fusion weight	0.539	0.281	0.009	0.009	0.009	0.008
σ	0.52	2.60	0.56	3.85	1.27	1.03
γ	1.24	0.52	0.70	0.50	1.14	0.79
Frequency /THz	0.7	0.8	0.9	1.0	1.1	1.2
Low-resolution original image						
Fusion weight	0.008	0.008	0.008	0.056	0.037	0.032
σ	1.44	0.93	4.78	5.24	3.38	1.73
γ	0.61	1.13	0.71	1.03	1.28	0.53
High-resolution result image and its 3D sample structure						

Fig. 8. Super-resolution result of DL-MFIF. (a) Sample 6; (b) Sample 7.

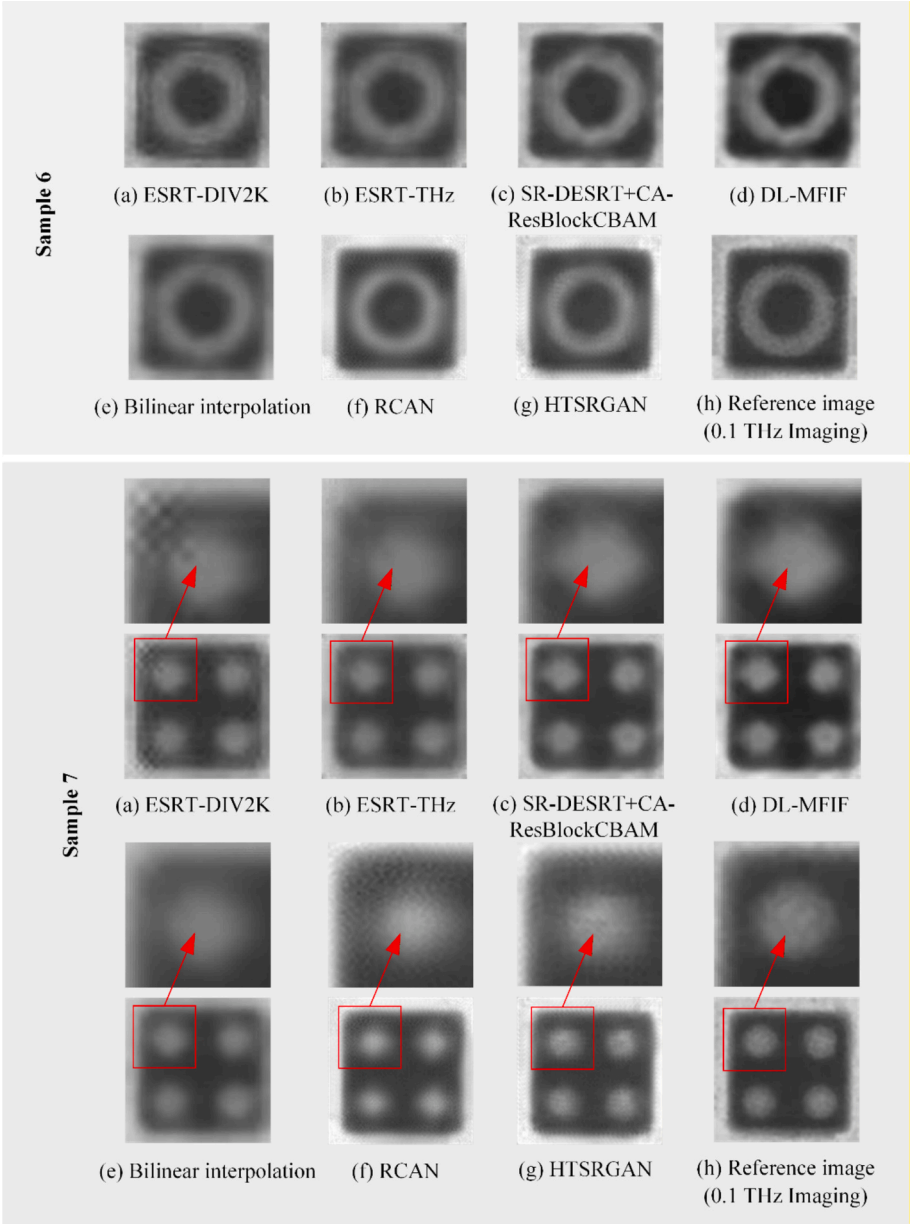


Fig. 9. Comparison of detail reconstruction in SR results by different methods.

Table 2
Average runtime for single-sample imaging.

Method	Times
SR-DESRT + CA-ResBlockCBAM	0.65 s
DL-MFIF	6.5 min
30 × 30 large-step point scanning	≈3 min
120 × 120 small-step point scanning	≈48 min

practical constraint, this study explored and demonstrated the feasibility of a super-resolution reconstruction and multi-frequency fusion framework designed for small-sample scenarios.

To mitigate the objective limitation of data scarcity, specific strategies were adopted at the algorithm design level. On the one hand, the proposed dual-branch network (SR-DESRT) incorporated a transfer learning mechanism. By freezing the weights of the optical branch pretrained on the DIV2K dataset during training, the model leveraged rich spatial texture priors learned from natural images to guide THz image reconstruction. As a result, the network's ability to recover edges and structural details did not rely solely on the limited THz training data, thereby helping to reduce the risk of overfitting to the geometric patterns of the resin specimens. On the other hand, the second stage of the proposed method (MFIF-GA) was an unsupervised optimization process driven by physically meaningful metrics, such as entropy and gradient information. Since its parameter optimization did not depend on the size of the training dataset, this stage exhibited stable behavior and favorable controllability under small-sample conditions.

In addition, a specimen-level data splitting strategy was adopted instead of random image-level splitting. Specifically, the specimens used in the test set did not appear during training, facilitating strict cross-specimen validation. Although this strategy was limited in statistical scale, it more closely reflected the engineering scenario of "new specimen testing" in practical THz inspection applications and facilitated the evaluation of model generalization to previously unseen samples. It should be noted that photosensitive resin specimens were selected as experimental objects due to their limited variability in texture and internal structure, which helped reduce uncertainty caused by material differences and enabled a controlled analysis of the proposed method. In

future work, more representative structures, such as composite components and metal welds, will be incorporated to further extend the applicability of the proposed framework.

The proposed method was primarily validated within the frequency range of 0.1–1.2 THz, which was determined based on a trade-off between signal-to-noise ratio and spatial resolution within the effective dynamic range of the employed THz-TDS system. The proposed framework does not rely on handcrafted features at specific frequencies; instead, it learns general spatial structural information from single-frequency images and performs multi-frequency fusion based on measurement-driven evaluation criteria. In principle, provided that sufficient signal quality is available, the method could potentially be extended to other frequency ranges. Nevertheless, it was acknowledged that changes in the frequency range may affect image contrast, penetration depth, and noise characteristics. If the bandwidth is too narrow, the complementary information available to the MFIF-GA module may be reduced, thereby limiting the fusion effectiveness. Conversely, including frequencies beyond the system's dynamic range, where signals are dominated by noise, may increase the optimization load on the GA to suppress invalid channels, which could adversely affect reconstruction performance.

Regarding noise robustness, although the dual-branch architecture integrated strong spatial priors from high-quality natural images to alleviate noise interference, the method implicitly assumed that structural edges remained at least partially discernible. Under extreme noise conditions, where structural information is completely obscured, the network may encounter difficulty in distinguishing high-frequency noise from genuine texture, potentially leading to reconstruction artifacts or excessive smoothing.

5. Conclusion

This study proposed a SR reconstruction method, DL-MFIF, for THz-TDS point-scan imaging, aiming to address key challenges in existing THz imaging systems, including insufficient spatial resolution, low imaging efficiency, and structural inconsistency across multi-frequency images. Unlike previous studies that primarily focused on single-frequency SR reconstruction, DL-MFIF integrates a dual-branch deep feature extraction network with a genetic-algorithm-based multi-frequency fusion strategy, enabling collaborative modeling and optimized fusion of spatial and frequency-domain information. Experimental results demonstrated that DL-MFIF outperformed conventional interpolation methods as well as deep learning–based SR approaches across both reference-based and no-reference image quality metrics. Specifically, DL-MFIF achieved clear improvements in variance, contrast, information entropy, and average gradient, indicating its superior capability in detail restoration and enhancement of spatial structural clarity. Moreover, the improvements in PSNR, SSIM, and LPIPS further verified that the proposed method can effectively balance pixel-level fidelity, structural consistency, and perceptual quality under small-sample conditions. The proposed method enabled high-quality THz-TDS image reconstruction under sparse sampling conditions. By reducing the dependence on dense point-scanning measurements, DL-MFIF significantly decreased imaging time costs and improved the operational efficiency of THz-TDS systems in practical engineering scenarios. Overall, DL-MFIF provides a feasible and extensible technical framework for achieving low-sampling, high-quality THz-TDS imaging, and offers effective technical support for the intelligent development of THz-TDS systems.

CRediT authorship contribution statement

Xinna Jiang: Writing – review & editing, Writing – original draft, Methodology, Validation. **Hongquan Jiang:** Conceptualization, Writing – review & editing. **Maojie Zhang:** Writing – review & editing, Software, Data curation. **Deyan Yang:** Writing – review & editing. **Yang Liu:** Conceptualization. **Jianmin Gao:** Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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